A Framework for Drift Detection and Adaptation in AI-driven Anomaly and Threat Detection Systems

Antonio Lara Gutiérrez[1], Carmen Fernández-Gago[1], and Jose A. Onieva[1]

Network, Information and Computer Security (NICS) Lab, University of Málaga
{alara, mcgago, onieva}@uma.es

Abstract. The dynamic and evolving nature of cybersecurity threats presents significant challenges to anomaly and threat detection systems, particularly those that rely on Artificial Intelligence (AI) as their detection engine. A key limitation of current AI models is their inability to adapt to concept drift, feature drift, and adversarial attacks, which degrade performance over time. This study introduces the Hybrid Drift Detection and Adaptation Framework (HDDAF), which is a multi-layered AI system that is specifically designed to mitigate concept drift, feature drift, and adversarial attacks in cybersecurity. HDDAF integrates Hoeffding drift detection, feature selection, adversarial training, and incremental learning, allowing it to dynamically adapt through a Mixed-Drift Handling Module, which balances fine-tuning and full retraining. Tested on the CIC-IDS2017 dataset, HDDAF achieved a macro F1 score exceeding 99.9%. Its hybrid architecture, which combines deep learning and ensemble methods, enhances detection accuracy and resilience, and an adaptive generative model anticipates drifts, ensuring scalability for real-time cybersecurity applications.

Keywords: Artificial Intelligence; Concept Drift, Feature Drift; Adversarial Attacks

1 Introduction

Intrusion Detection Systems (IDS) emerged from the increasing demand for proactive security measures to monitor system-level activities and detect unauthorised or malicious behaviour within a system or network. James P. Anderson's foundational work in the 1970s introduced the concept of monitoring user actions for suspicious activities, thus laying the groundwork for contemporary IDS practices. Building on this foundation, Dorothy E. Denning proposed a formal model in the 1980s that emphasised real-time anomaly detection and audit data analysis as critical security strategies.

As cyberattacks have become increasingly complex and sophisticated, there is a growing need for more dynamic IDSs capable of handling diverse, large-scale, and real-time scenarios. Early IDS solutions primarily relied on static rule-based

engines and statistical analysis, which quickly proved inadequate for scalability and for detecting novel threats. The integration of Artificial Intelligence (AI) techniques, including Machine Learning (ML) and Deep Learning (DL), has introduced a paradigm shift in the capabilities of IDS.

ML models are valued for their interpretability; however, they struggle with scalability and high-dimensional data. In contrast, DL models excel at capturing complex temporal and spatial patterns but are computationally demanding. Hybrid models that combine ML and DL techniques have emerged as promising alternatives, offering a balance between performance and computational efficiency.

Despite these advancements, existing AI models face critical challenges when deployed in real-world cybersecurity environments. Concept drift [9], where data distributions evolve, and feature drift [3], where feature importance changes and degrades model performance, may compromise future model outputs. In addition, adversarial attacks exploit vulnerabilities in AI systems, which reduces their robustness and reliability. Developing adaptive methods capable of detecting and mitigating drift while maintaining system performance can address these issues [20].

Many existing frameworks fail to provide a comprehensive approach that dynamically handles both concept and feature drift while maintaining system resilience and long-term adaptability.

To address these challenges, this paper introduces the Hybrid Drift Detection and Adaptation Framework (HDDAF), which is an enhanced framework designed for AI-driven anomaly and threat detection systems. HDDAF integrates a multilayered hybrid AI architecture with active adaptation modules to maintain detection performance in dynamic threat landscapes. The primary contributions of this work are:

1. A mixed-drift handling module that dynamically identifies and responds to concept and feature drifts.
2. An adaptive generative model leverages GANs to pre-train the model for future predictive drift.
3. Robustness against adversarial attacks through data poisoning using adversarial training techniques.
4. Experimental evaluation shows that the proposed method outperforms existing approaches.

This article is structured as follows. Section 2 reviews previous works addressing anomaly detection, drift management, and adaptive AI systems. Section 3 details the proposed methodology, including its architectural design and components. Section 4 presents the results of the experimental evaluation and discusses the insights extracted from the development. Finally, Section 5 concludes the paper, outlining future research directions to further enhance AI-driven anomaly detection systems.

2 Related Work

The challenge of managing evolving data distributions in AI-driven anomaly and threat detection has become a critical focus in recent years. Numerous frameworks and methodologies have been introduced to enhance the adaptability, robustness, and overall performance of such systems. Table 1 provides a comparative overview of various studies, highlighting the diverse areas they address and their contributions to the field.

Reference	Approach	Concept Drift	Feature Drift	Feature Engineering	Adversarial Attacks	Dataset(s) Used
Zhang et al. (2024) [21]	HCDD-SHAP (Explainability-based multilayer drift detection)	✓	✗	✓	✗	Synthetic, Real-World
Mulimani et al. (2024) [10]	Online LightGBM with Sliding Window and Incremental Learning	✓	✗	✓	✗	Electricity, Spam, MixedAbrupt
Aguiar and Cano (2023) [2]	Meta-Learning for Concept Drift Detection	✓	✗	✓	✗	10 Real-World + 18 Synthetic
Noori et al. (2023) [12]	Dynamic Feature Aware GP Ensemble (DFA-GPE)	✓	✓	✓	✗	HIKARI 2021, TON_IoT 2020
Wisanwanichthan et al. (2021) [18]	Double-Layered Hybrid Approach (Naive Bayes + SVM)	✓	✓	✓	✗	NSL-KDD
ElDahshan et al. (2022) [5]	Meta-Heuristic Optimisation-Based Hierarchical IDS	✓	✓	✓	✗	UNSW-NB15, CICIDS2017
Halbouni et al. (2022) [7]	CNN-LSTM Hybrid Deep Neural Network	✓	✓	✓	✗	CIC-IDS2017, UNSW-NB15, WSN-DS
Zhao et al. (2022) [22]	Hybrid IDS with Feature Selection and Weighted Stacking	✓	✓	✓	✗	NSL-KDD, CSE-CIC-IDS2018
Sheikhi et al. (2022) [16]	PSOGWO-Optimised BP Neural Network	✓	✓	✓	✗	NSL-KDD
Harwahyu et al. (2024) [8]	Three-layer hybrid learning (LSTM + RF)	✓	✓	✓	✗	CSE-CIC-IDS2018
Turukmane et al. (2024) [17]	M-MultiSVM with Advanced Feature Selection	✓	✓	✓	✗	CSE-CIC-IDS2018, UNSW-NB15
Our Proposal	Hybrid Drift Detection and Adaptation Framework (HDDAF)	✓	✓	✓	✓	Real (CICIDS2017), synthetic and adversarial

Table 1. Comparison of Related Works on Drift Detection and Adaptation Frameworks.

A key advancement in this field is the integration of explainability and dynamic adaptability into drift detection frameworks. Zhang et al. (2024) [21] pioneered the HCDD-SHAP method, a multilayer drift detection approach that combines model explainability with precise drift identification. The framework sequentially detects drift through error rate variations, validates it using SHAP values to assess feature contributions, and identifies the drift onset point with a Kneedle algorithm. This layered design enhances interpretability and precision, making it particularly effective in contexts requiring transparency.

Complementing Zhang et al.'s work, Mulimani et al. (2024) [10] introduced an online drift detection and adaptation framework leveraging the Light Gradient Boosting Machine (LGBM). Their approach integrates incremental learning, ensemble methods, and a sliding window technique to detect and adapt to

concept drift in real-time data streams. The experiments demonstrated superior accuracy and adaptability compared to traditional models. While Zhang et al. emphasised explainability, Mulimani et al.'s framework excelled in dynamic adaptation, making it better suited for real-time applications.

Aguiar and Cano (2023) [2] proposed a meta-learning framework that dynamically selects drift detectors based on statistical and temporal features. By tailoring detection mechanisms to the characteristics of evolving data, the proposed system highlights the importance of flexibility when addressing diverse scenarios.

To address feature drift explicitly, Noori et al. (2023) [12] introduced the Dynamic Feature-Aware GP Ensemble (DFA-GPE), which integrates a Variable Length Multi-Objective Particle Swarm Optimisation (VLMO-PSO) technique. This framework dynamically manages feature importance, achieving memory efficiency and high accuracy in high-dimensional datasets like HIKARI [1] and TON-IoT 2020 [2]. Unlike Zhang et al. and Mulimani et al., who focused on concept drift and adaptation, Noori et al. excelled in environments where feature importance evolves over time, which is a critical factor in handling high-dimensional data.

Expanding the concept to hybrid model designs, Wisanwanichthan et al. (2021) [18] combined Naive Bayes and SVM classifiers into a double-layered approach. Validated on the NSL-KDD [3] dataset, this system effectively handled various attack patterns, demonstrating the value of targeted designs for specific applications.

Incorporating optimisation techniques, ElDahshan et al. (2022) [5] used metaheuristic algorithms to fine-tune hierarchical intrusion detection systems. The proposed method not only improved detection rates but also demonstrated the potential of hybrid optimisation methods to address complex data distributions.

Deep learning-based methodologies also show promise. Halbouni et al. (2022) [7] combined CNN and LSTM architectures to capture spatial and temporal patterns in network traffic. Their hybrid model achieved notable success in detecting anomalies across datasets such as CIC-IDS2017 [4] and UNSW-NB15 [5], underscoring the efficacy of neural networks in cybersecurity contexts.

Zhao et al. (2022) [22] enhanced performance through a hybrid approach that integrated feature selection and weighted stacking. By prioritising feature relevance, their model achieved significant improvements on NSL-KDD and CSE-CIC-IDS2018 [6], illustrating the importance of preprocessing in addressing complex data challenges.

Based on these developments, Harwahyu et al. (2024)[8] and Turukmane et al. (2024)[17] explored hybrid learning and advanced feature selection techniques,

[1] HIKARI dataset available at `https://zenodo.org/record/5199540`

[2] TON-IoT datasets available at `https://research.unsw.edu.au/projects/toniot-datasets`

[3] NSL-KDD dataset available at `https://web.archive.org/web/20150205070216/http://nsl.cs.unb.ca/NSL-KDD/`

[4] CIC-IDS2017 dataset available at `https://www.unb.ca/cic/datasets/ids-2017.html`

[5] UNSW-NB15 dataset available at `https://research.unsw.edu.au/projects/unsw-nb15-dataset`

[6] CSE-CIC-IDS2018 dataset available at `https://www.unb.ca/cic/datasets/ids-2018.html`

respectively. Harwahyu et al. proposed a three-layer hybrid learning model that combined LSTM and random forest, whereas Turukmane et al. employed an M-MultiSVM framework with advanced feature selection strategies. Both approaches further emphasise the value of integrating diverse machine learning techniques to enhance the detection and adaptation of drift.

A comparative analysis of these studies revealed common threads in the pursuit of enhanced drift detection and adaptation frameworks. Many authors have advocated continuous model updates and dynamic mechanisms for drift detection, reflecting the practical constraints of real-world applications. The proposed framework synthesises these insights with the aim of extending the capabilities of existing models, thereby offering a comprehensive solution to the multifaceted problem of drift in dynamic environments.

3 Hybrid Drift Detection and Adaptation Framework (HDDAF)

In this paper, we have presented a scalable and adaptive framework for anomaly and threat detection called the Hybrid Drift Detection and Adaptation Framework (HDDAF), is presented. The proposed framework is designed to address the evolving challenges posed by concept drift, feature drift, and adversarial attacks in AI-driven systems. The system integrates multi-layered hybrid AI approaches with advanced preprocessing, dynamic drift adaptation mechanisms, and adversarial training to ensure long-term resilience and maintainability over time. The proposed approach is presented in Figure 1.

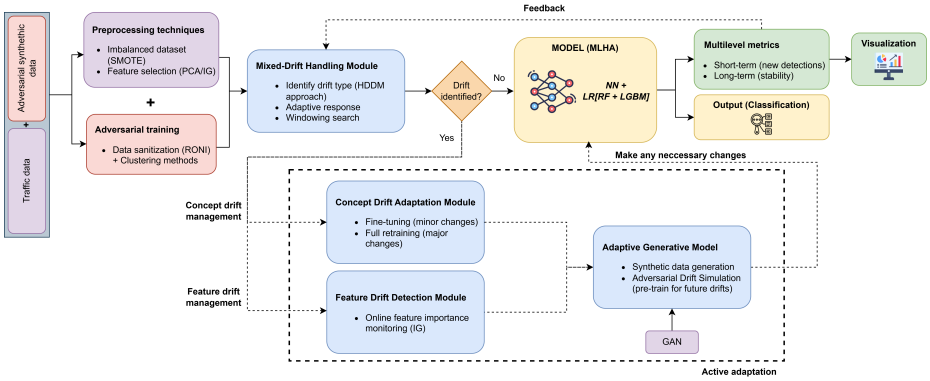


Fig. 1. Hybrid Drift Detection and Adaptation Framework (HDDAF)

3.1 Dataset

The dataset that represents the real traffic data in the proposed framework is CIC-IDS2017 [15], developed by the Canadian Institute for Cybersecurity using a university network with different subnets and mixed real traffic. The dataset includes 14 types of labelled individual attacks, which we have grouped in 8 families, namely Bot, Brute Force, DDoS, DoS, Heartbleed, Infiltration, Port Scan, and Web Attack. It is widely used for benchmarking intrusion detection systems due to its comprehensive coverage of network behaviours and threats. The dataset comprises 85 features representing flow-level statistics, with each row representing a data flow.

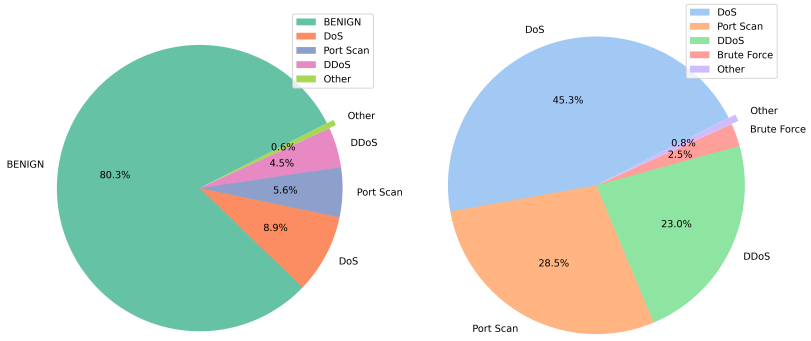


Fig. 2. Attack Distribution in CIC-IDS2017: Overall Traffic (left) and Attack Types Breakdown (right)

An initial Exploratory Data Analysis (EDA) was conducted to better understand the characteristics of the dataset. Figure 2 shows the distribution of labels and the proper distribution of attacks found, categorised into 8 different types of attacks. Notably, class imbalance was observed, with benign traffic dominating over attack instances, highlighting the need for balancing techniques during the model training phase.

Table 2 highlights the features with the highest positive correlation against the target variable, providing valuable insights into the underlying patterns that distinguish different attack behaviours. Features such as "Bwd Packet Length Std" and "Bwd Packet Length Max", with correlations of 0.33 and 0.32, respectively, indicate a strong relationship with the occurrence of specific attack types. These high correlations suggest that changes in backward packet length metrics are particularly influential in identifying attacks, possibly due to their sensitivity to data flow pattern anomalies.

Feature	Correlation	Feature	Correlation
Bwd Packet Length Std	0.33	Idle Max	0.29
Bwd Packet Length Max	0.32	Flow IAT Max	0.28
Bwd Packet Length Mean	0.31	Fwd IAT Max	0.28
Fwd IAT Std	0.31	Idle Mean	0.28
Avg Bwd Segment Size	0.31	Idle Min	0.28
Packet Length Std	0.30	PSH Flag Count	0.27
Max Packet Length	0.29	Packet Length Mean	0.25
Packet Length Variance	0.29	Average Packet Size	0.25

Table 2. Features with Positive Correlation with Attack Number

3.2 Pre-training techniques

Different preprocessing techniques were applied to the raw dataset. For example, feature selection is implemented after minor adjustments, such as null filling, duplicate removal, or numeric filtering, to enhance the computational load and reduce complexity. The following two approaches were evaluated:

- **Principal Component Analysis (PCA) [1]**: A dimensionality reduction method that transforms features into a smaller set of uncorrelated components (called PCX). In this case, 98.58% of the information was retained, leaving the dataset with only 36 features and the target variable. However, interpretability is no longer possible as the variables are now transformed into uncorrelated components (PCs), which leaves us with no information about model decisions, crucial for debugging, compliance, and making informed decisions in critical decisions or audits.
- **Information Gain (IG) [6]**: Measures the reduction in entropy relative to the target variable when a single feature is used for classification. Essentially, it quantifies how well a feature separated the classes in a dataset. Then, we can select some features that we want to keep for further analysis, arranged by their IG results. To manage the number of features that will be selected, we conducted a test to evaluate the classification performance of the datasets by including a successive number of features, ordered by their IG value. Figure 3 shows a moving average of the accuracy and F1 scores obtained depending on the number of selected features. The optimal value that will be chosen for future experiments will be 32.

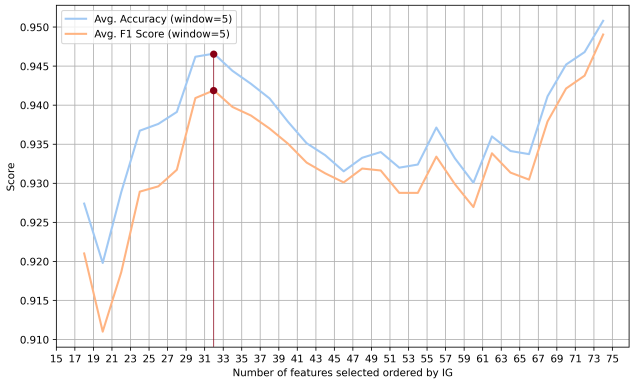


Fig. 3. Rolling Average of Accuracy and F1 Score vs. Number of Selected Features Ordered by IG values.

Given the previously identified class imbalance, the synthetic minority oversampling technique (SMOTE) was applied [4]. SMOTE can generate synthetic examples for minority classes to balance the dataset. In this case, as seen in Figure 2, these classes correspond to the rarest attacks.

To strengthen the model's resilience against adversarial attacks, a hybrid adversarial training technique was developed. Figure 5 summarises the progression of results obtained during the development of the proposed technique, which combines:

- **Data Sanitisation with RONI (Reject on Negative Impact) [11]**: RONI takes each sample, removes it from the dataset, and compares the validation accuracy obtained with a baseline reference value. If the accuracy was improved, the sample was flagged as 'Rejected'. While this filtering helps filter out harmful or misleading data, it also leads to the removal of rare but important samples, which reduces model generalizability. Additionally, the method's reliance on baseline accuracy makes it sensitive to noise and potential biases in the reference model. The overuse of RONI could result in a sanitised but overly selective dataset, limiting the model's adaptability to diverse real-world scenarios.
- **Clustering methods**: Used to identify groups of data points, where outliers are separated into separate clusters.
- **Hybrid Adversarial Training Technique**: This technique identifies possible outliers using clustering methods (specifically, with the KMeans [19] algorithm) and then applies RONI individually to suspicious data points. The proposed approach reduces computational costs by limiting RONI evaluations to high-risk samples rather than the entire dataset. The proposed method also mitigates the issue of removing rare but valuable samples by

preserving samples that align with recognised data clusters. Furthermore, by combining clustering with RONI, the proposed method becomes less sensitive to noise and potential biases in baseline accuracy, thereby ensuring a more balanced and adaptable model. Figure 4 shows a graphical representation of how this technique works and how it affects the data, which will be analysed by the model in future steps.

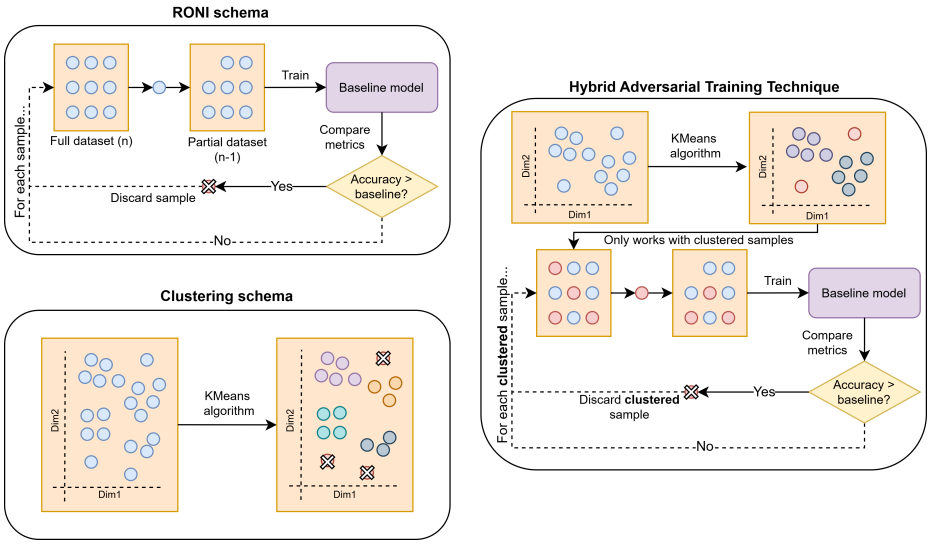


Fig. 4. Graphical Representation of RONI, Clustering and Hybrid Adversarial Training Technique Working Principles

When trying to obtain the maximum precision, RONI excels because it manages to obtain a low number of false positives (high precision), but the execution time can be extensive. However, clustering methods often identify many false positives, resulting in low precision but high sensitivity. To achieve balance in all these aspects, the proposed hybrid technique works seamlessly in all these aspects, as shown in the comparison plotted on Figure 5.

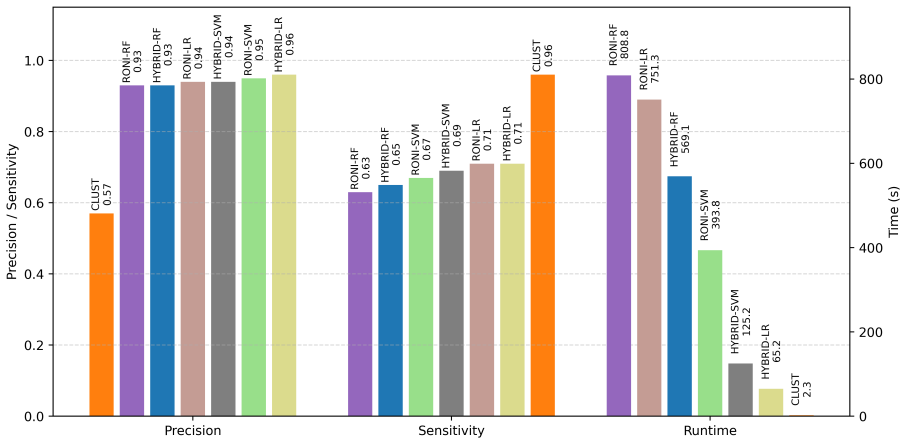


Fig. 5. Precision, Sensitivity, and Runtime Comparison of Selected Techniques for Data Sanitisation

3.3 Hybrid Model

The HDDAF employs a multilayered hybrid AI architecture to maintain robustness and high detection accuracy under evolving conditions. In addition, we implement measures to retrain the model and maintain a high accuracy value.

The hybrid model, depicted in Figure 6, combines a neural network (NN) comprising two hidden layers with an ensemble of Random Forest (RF) and LightGBM (LGBM), finalised with a Logistic Regression (LR) meta-classifier. The NN serves as the initial detection layer, extracting features, and performing binary classification to distinguish between benign traffic and potential anomalies. The deep architecture of the proposed method enables the detection of complex patterns, while dropout regularisation helps reduce overfitting. In comparison with other approaches, binary classification using an NN demonstrated high precision and low sensitivity, making it highly suitable for this task. Then, the ensemble layer combines RF and LGBM, where RF provides feature importance and variance reduction, and LGBM offers gradient boosting with efficient handling of large datasets and imbalanced traffic patterns.

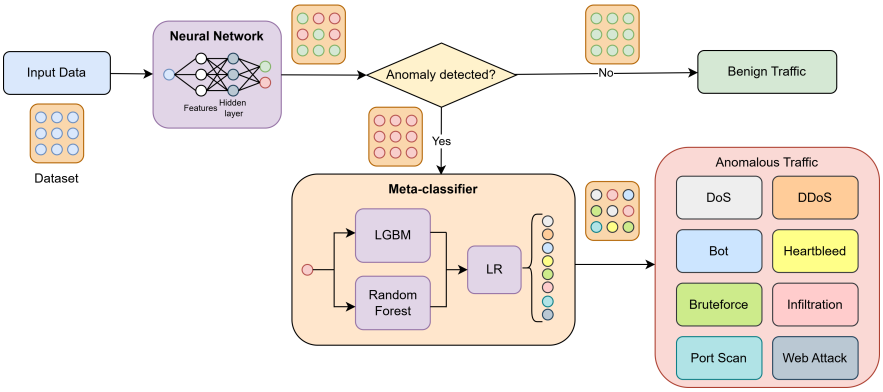


Fig. 6. Representation of the Hybrid Model Proposed for Classification

The final decision layer employs a logistic regression meta-classifier to aggregate predictions from both RF and LGBM, combining their outputs in a weighted manner to enhance overall accuracy. The use of a Logistic Regression meta-classifier for the final aggregation ensures a unified and optimised output by combining predictions in a weighted manner. The specifications of the implementations are as detailed in Table 3. The selected hyperparameters are obtained from a cross-validated grid-search over a parameter grid for each classifier used.

The overall architecture is designed to address the key challenges of modern intrusion detection systems, including scalability, adaptability to drifts, and high accuracy under real-time constraints. The integration of deep learning, ensemble methods, and meta-learning ensures that the proposed framework achieves a balance between robustness and efficiency, making it suitable for large-scale and continuously evolving data streams.

Model	**Parameters / Configuration**
Neural Network	**Architecture:** Dense(128, ReLU) → Dropout(50%) → Dense(64, ReLU) → Dropout(50%) → Dense(2, Softmax) **Compilation:** loss = `categorical_crossentropy`, optimizer = `adam` (lr = 0.001), metric: `accuracy` **Training:** 50 epochs, batch size = 256
Random Forest	`n_estimators` = 200, `max_depth` = 15, `criterion` = `gini`, `max_features` = `sqrt`
LightGBM	`min_gain_to_split` = 0.1, `min_child_samples` = 20, `min_data_in_leaf` = 30, `num_leaves` = 40, `max_depth` = 15, `learning_rate` = 0.05, `feature_fraction` = 0.8, `bagging_fraction` = 0.8, `bagging_freq` = 5
Logistic Regression	`max_iter` = 20000, `C` = 1.0, `penalty` = `l2`, `solver` = `lbfgs`

Table 3. Hyperparameters and Configurations for the Hybrid Model Components.

3.4 Drift Handling

The drift handling implemented in this framework is based on an active adaptation module, which is designed to dynamically identify, monitor, and respond to data distribution shifts, which is a common challenge in such systems. The component initially works using a Hoeffding Drift Detection method [13] that continuously monitors the incoming data for signs of drift, as depicted in the Mixed-Drift Handling Module in Figure 1. This statistical test can detect gradual and sudden changes in the data distribution. When drift is detected, the system activates the Concept Drift Adaptation Module and/or the Feature Drift Detection Modules. The first parameter can now decide whether fine-tuning is applied or total retraining of the main model is due, depending on the severity of the drift. The second method computes the new IG values of the features and selects which ones are responsible for the new retraining of the model.

When full retraining is performed, an Adaptive Generative Model is triggered. The model is responsible for simulating new drifts using a Generative Adversarial Network (GAN) [14] to check the resilience of the new model and future retraining, if necessary. Algorithm 1 explain how this is performed using pseudocode.

Algorithm 1 Drift Handling Algorithm with an Active Adaptation Module

Require: Incoming data stream D
Ensure: Updated model M
Initialise model M and Mixed-Drift Handling Module
while True **do**
    Observe new data batch B from D
    Apply Hoeffding Drift Detection to B
    if Drift Detected **then**
        if Concept Drift Detected **then**
            Activate Concept Drift Adaptation Module
            Compute severity of drift
            if Severe Drift **then**
                Perform full retraining of model M
            else
                Apply fine-tuning to model M
            end if
        end if
        if Feature Drift Detected **then**
            Activate Feature Drift Detection Module
            Compute new Information Gain (IG) values for features
            Select features responsible for drift
            Retrain model M with the updated feature set
        end if
        if Full retraining executed **then**
            Trigger Adaptive Generative Model using GAN
            Simulate new drifts and test model resilience
            Perform additional retraining if required.
        end if
    end if
    Update model M with B
end while

4 Results and discussion

This section presents the results obtained from the experimental evaluation of the proposed framework. All experiments were conducted on a workstation equipped with an AMD Ryzen 7 5800H CPU (8 cores, 16 threads), 16 GB DDR4 RAM, and an NVIDIA RTX 3060 GPU running Ubuntu 24.04 LTS. The software environment included Python 3.12.8 using key libraries such as scikit-learn (1.6.0), TensorFlow (2.18.0) with Keras (3.8.0), and NumPy (2.0.2), among others.

4.1 Model evaluation

The first part of the experiment involved the hybrid multi-layered model designed, which we tested across several scenarios depending on feature selection, preprocessing for imbalance adaptation, and the type of model used (NN, RF, LGBM, ensemble method of RF+LGBM and our hybrid method), as shown in Tables 4 and 5. The metrics evaluated are the accuracy of the model as the closeness between the predicted and actual classes, the macro F1 score as the arithmetic mean of all the per-class F1 scores, and the time as the time consumed by the OS to train each model with the provided batch of data.

		RF			**k-NN**			**LGBM**			**NN**		
		Acc	**F1**	**Time (s)**	**Acc**	**F1**	**Time (s)**	**Acc**	**F1**	**Time (s)**	**Acc**	**F1**	**Time (s)**
NO-SMOTE	PCA (37)	.9735	.5439	41.70	.9979	.8157	1.58	.7933	.0983	5.72	.9993	.9744	25.15
	Top 32 IG	.9912	.6312	3.16	.9642	.6161	1.59	.9341	.4080	5.81	.7995	.1269	26.02
SMOTE	PCA (37)	.9304	.9325	37.11	**.9930**	**.9930**	**1.62**	**.9992**	**.9992**	**7.45**	**.9997**	**.9997**	**23.55**
	Top 32 IG	**.9845**	**.9840**	**6.04**	.9632	.9627	1.60	.9980	.9980	9.58	.1433	.0358	24.58

Table 4. Performance of **RF**, **k-NN**, **LGBM**, and **NN** under different feature selection, dimensionality reduction, and oversampling settings.

		XGBoost			**SVM**			**LR[RF+LGBM]**			**NN+LR[RF+LGBM]**		
		Acc	**F1**	**Time (s)**	**Acc**	**F1**	**Time (s)**	**Acc**	**F1**	**Time (s)**	**Acc**	**F1**	**Time (s)**
NO-SMOTE	PCA (37)	.9987	.8850	1.03	.9988	.8314	3.22	.8512	.8512	267.15	.9420	.9380	312.95
	Top 32 IG	.9321	.8619	0.75	.8727	.3333	21.38	.9321	.9211	57.43	.7699	.7699	45.12
SMOTE	PCA (37)	**.9972**	**.9972**	**1.12**	**.9985**	**.9985**	**3.83**	.9995	.9995	235.21	.9999	.9999	285.40
	Top 32 IG	.9921	.9921	0.89	.5970	.5961	73.24	**.9996**	**.9996**	**61.80**	**.9999**	**.9999**	**62.50**

Table 5. Performance of **XGBoost**, **SVM**, **LR[RF+LGBM]**, and **NN+LR[RF+LGBM]** under different feature selection, dimensionality reduction, and oversampling settings.

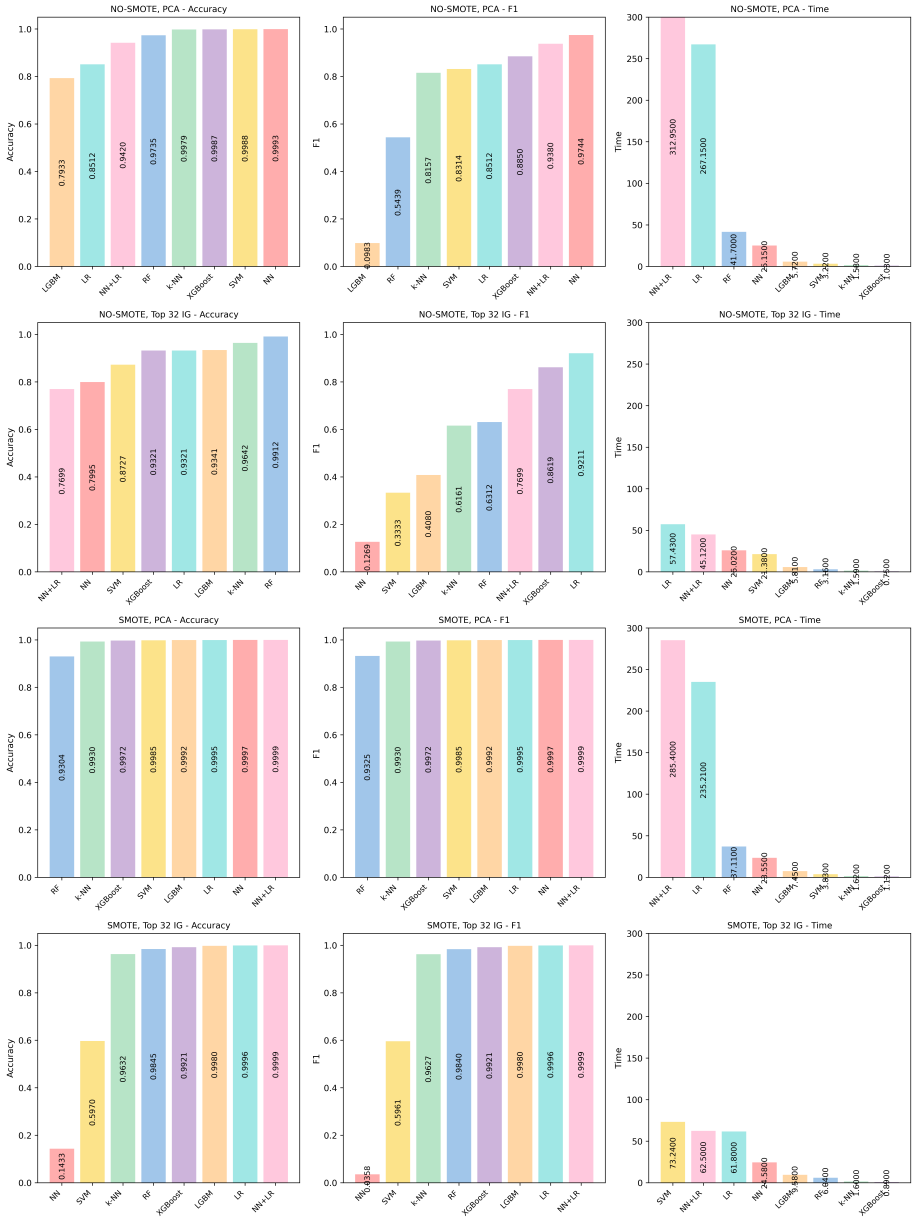


Fig. 7. Graphical Comparison of Results between all Models Tested

A consistent observation across our tests is that incorporating SMOTE generally enhances performance. Similarly, the choice of feature selection method

plays a critical role: while PCA tended to provide stable results across models, the Top 32 Information Gain approach yields mixed outcomes, illustrating a trade-off between dimensionality reduction and the preservation of discriminative features.

When examining individual models, clear trade-offs emerge. Simpler models like k-NN model, offer the advantage of fast training times; however, they can lag in terms of classification robustness. Conversely, models such as RF and LGBM deliver high accuracy and F1 scores in many configurations; however, their performance is sensitive to the feature selection method employed. In particular, the standalone Neural Network demonstrated exceptional results in some settings (e.g., high accuracy with PCA) but suffered when faced with less optimal feature configurations.

Balancing these considerations, the final model selected for the proposed framework is the NN+LR[RF+LGBM] ensemble. This hybrid approach leverages the deep learning model's ability to capture complex patterns and the ensemble's strength in aggregating diverse predictive insights. Although NN+LR[RF+LGBM] can require longer training times in certain configurations, its consistently superior accuracy and macro F1 score—especially when combined with SMOTE—make it the most robust option for handling imbalanced data and variable feature spaces. In addition, the selection of features based on the IG was the most optimal in terms of accuracy and computational time.

4.2 Framework adaptability evaluation

The second part of the experiment aimed to evaluate the performance of the HDDAF as a whole under various drift scenarios and adversarial attack conditions. To assess the overall performance of the HDDAF framework under dynamic and challenging conditions, we designed a series of experiments to simulate various drift scenarios and adversarial attack conditions. In these experiments, the hybrid NN+LR[RF+LGBM] model was deployed within the full framework, while other modules remain unchanged from the previous section. The evaluation focused on the following three key objectives:

1. **Drift Detection and Recovery**: Measuring how quickly and effectively the system identifies and adapts to changes in the data, measured by the number of delayed batches until detection.
2. **Retraining Efficiency**: Comparing the computational cost and adaptation effectiveness of fine-tuning versus full retraining methods, measured using the retraining time and accuracy.
3. **Adversarial Robustness**: Evaluating the framework resilience when confronted with adversarial samples designed to poison the data, measured with the number of adversarial samples correctly detected and removed.

To simulate these conditions, data is streamed into the system in batches ranging from 2000 to 5000 samples. Each batch initially follows a baseline distribution, after which specific drift or adversarial scenarios are introduced at predetermined intervals.

Concept Drift scenarios mimic changes in the underlying relationship between features and labels:

- **Gradual Drift**: The decision boundary shifts slowly over several batches, indicating a gradual evolution of the concept. This was implemented by applying a 5-10% incremental change in feature-label correlation across 10-20 batches, gradually modifying the target distribution while maintaining continuity.
- **Sudden Drift**: A rapid, abrupt change is introduced at a specific batch to simulate an unexpected alterations in the data generation process. At a fixed batch index, samples are replaced with a new distribution with different statistical properties, which immediately shifts the decision boundaries.
- **Recurring Drift**: The data distribution alternates between different regimes in a cyclical pattern, reflecting environments where previously encountered concepts reoccur. This is achieved by cycling through previously observed distributions every 20-30 batches, simulating seasonal variations or periodic changes in user behaviour.
- **Incremental Drift**: Small, consistent modifications are applied over successive batches, leading to a steady, cumulative shift in the concept. This is introduced by applying slight, continuous transformations to the numerical features across all incoming batches, a small additive Gaussian noise in our case.

Feature Drift scenarios focus on changes at the feature level:

- **Feature Removal**: Specific features are removed from the data stream at set intervals, simulating scenarios such as failures or missing data. This process is executed by randomly masking or entirely dropping certain features from input data every 5-10 batches.
- **Feature Addition**: New features are introduced into the stream to simulate the integration of additional data sources. This process is performed by appending new synthetic or engineered features based on the transformations of existing features every 10-20 batches.
- **Correlation Change**: The inter-feature relationships were altered, affecting the overall structure and correlation patterns in the dataset. This is achieved by dynamically modifying the correlation coefficients between key features or reshuffling feature dependencies across different batches.

To test adversarial robustness, we simulate attacks by manually perturbing a subset of the samples:

- **Adversarial Samples**: Adversarial samples are created by shifting or corrupting feature values, thereby attempting to mislead the model through data poisoning. This is achieved by injecting mislabelled samples into the dataset every 5-20 batches.

Each scenario was designed to reflect realistic conditions that a deployed system may face. By streaming the data in controlled batches and introducing these scenarios at predefined intervals, we can accurately monitor performance changes. The framework response is evaluated in terms of detection delay, accuracy before and after drift events, retraining time, and ability to withstand adversarial manipulation.

Table 6 provides an overview of the experimental results, summarising key metrics such as detection delays, pre- and post-drift accuracies, retraining methods, and times, as well as the measured adversarial robustness. Figure 8 represents box plots of model accuracies before and after drift for seven distinct scenarios, illustrating the range of performance for each scenario.

Drift Type	**Drift Scenario**	**Detection Delay (Batches)**	**Accuracy Before Drift (%)**	**Accuracy After Drift (%)**	**Retraining Method**	**Retraining Time (s)**
Concept Drift	Gradual	3	99.95	96.50	Fine-tuning	12.20
Concept Drift	Sudden	2	99.95	97.75	Full Retraining	75.62
Concept Drift	Recurring	3	99.80	98.22	Fine-tuning	13.50
Concept Drift	Incremental	4	99.95	96.80	Fine-tuning	12.86
Feature Drift	Removal	1	99.96	94.02	Full Retraining	74.10
Feature Drift	Addition	2	99.78	96.15	Fine-tuning	18.05
Feature Drift	Corr. Change	3	99.89	97.68	Full Retraining	76.78

Table 6. Drift Robustness Experimental Results

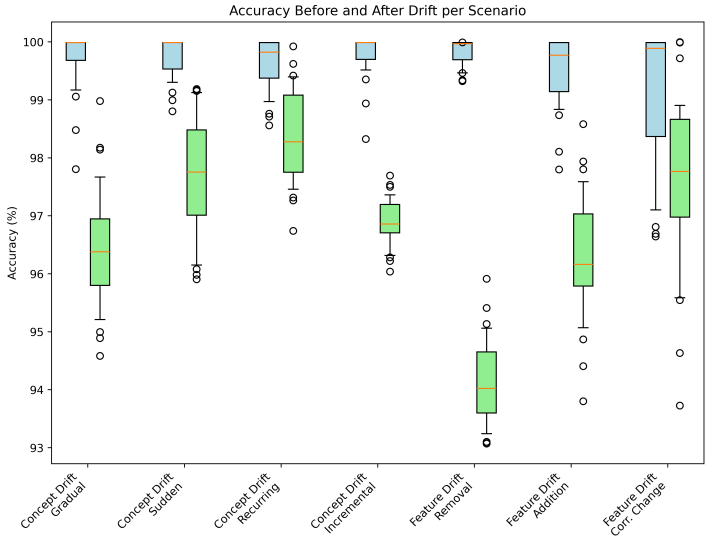


Fig. 8. Boxplot Representation for Tests Performed in Table 6

Figure 9 illustrates the effect of increasing poisoning percentages on model accuracy and Mean Squared Error (MSE), highlighting both the vulnerability of the baseline classifier and the effectiveness of the hybrid defence model explained before, consisting of a Hybrid Adversarial Training Technique.

As the poisoning percentage increases, the adversarial accuracy declines steadily, whereas the adversarial MSE rises sharply, demonstrating the growing distortion introduced by poisoned data. However, after applying RONI and clustering, the model maintained near-baseline accuracy and a significantly lower MSE, indicating successful mitigation of adversarial interference.

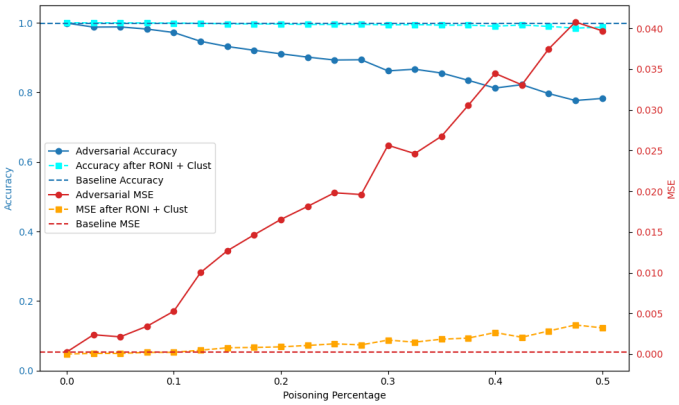


Fig. 9. Evolution of Performance Metrics for a Poisoning Attack Scenario

4.3 Key Insights

The key insights derived from these results are as follows:

- PCA-based models provide better accuracy but incur higher computational costs because of the complexity of the reduction. In addition, explainability is compromised as the features are transformed.
- Methods using IG information were more efficient but underperformed in comparison to PCA approaches.
- Combining models (e.g., through LR[RF+LGBM] and NN+LR[RF+LGBM]) provides a balanced solution that enhances both accuracy and robustness. Although these ensembles require greater computational resources, they consistently outperform individual models in handling class imbalances and feature drift.
- The Mixed-Drift Handling Module is effective in balancing model stability and efficiency. By using Hoeffding Drift Detection, the system dynamically decides between fine-tuning and full retraining based on drift severity, ensuring minimal disruption to ongoing operations. The accuracies obtained after drift consistently similar to previous ones.

- Hybrid adversarial training combines clustering-based outlier detection and RONI sanitisation, achieving a good balance in terms of accuracy and MSE when tested on adversarial poisoning attack scenarios.
- The modular design of the framework ensures scalability and adaptability. The ability of the proposed method to integrate incremental updates, semi-supervised learning, and cross-domain applications makes it a future-proof solution for real-world anomaly detection.
- In comparison to other methods, HDDAF offers distinct advantages. It surpasses approaches such as HCDD-SHAP[21] and DFA-GPE [12] in both concept and feature drift handling. The proposed design ensures scalability and adaptability, making it suitable for real-time deployment in resource-limited environments.
- Practical applicability was validated using the CIC-IDS2017 dataset, which mirrors real-world network conditions.

5 Conclusion

This study demonstrates the effectiveness of the Hybrid Drift Detection and Adaptation Framework (HDDAF) in addressing concept drift, feature drift, and resilience in AI-driven threat and anomaly detection systems. By integrating deep learning with ensemble machine learning, HDDAF achieves superior adaptability compared to traditional approaches. The Mixed-Drift Handling Module dynamically determines whether fine-tuning or full retraining is necessary to balance stability and efficiency. The framework's modular and scalable design ensures its practicality for production deployment, as validated by testing on the CIC-IDS2017 dataset.

Unlike conventional models, which tend to struggle with evolving data distributions, the proposed framework proactively detects and mitigates drifts using advanced adaptive learning techniques. Notably, HDDAF outperformed benchmark models found in literature such as HCDD-SHAP and DFA-GPE, which exhibit lower resilience to evolving data distributions. The CNN-LSTM-based IDS, a model recognised for its efficiency in handling time sequences, performed well on the same dataset with an F1-score of 97.7%. However, HDDAF maintained a macro F1 score above 99.9%, ensuring higher precision and fewer false positives.

The experimental results confirm that HDDAF adaptability and robustness stem from both its hybrid ensemble structure and the strategic use of feature selection. As discussed, PCA often delivered strong performance; however, information gain (IG)-based selection offered a superior balance between accuracy and computational efficiency across different models. In addition, incorporating SMOTE consistently improved classification outcomes, which highlights the importance of managing class imbalance. In particular, the NN+LR[RF+LGBM] hybrid provided consistently high accuracy and macro F1 scores under varying drift and adversarial scenarios, demonstrating the framework capacity for real-world resilience and adaptability.

One of HDDAF most valuable contributions is its ability to handle feature drift. DFA-GPE, for example, focused extensively on high-dimensional feature management but lacked a generative modelling component, making it vulnerable to unpredictable data variations. HDDAF adaptive generative modelling allows it to anticipate drift rather than simply react to it, significantly improving long-term stability. However, despite its strengths, HDDAF is not without trade-offs. The computational expense introduced by adaptive retraining and adversarial training, while efficient, still presents challenges for real-time edge deployments where processing power is limited. In addition, adversarial training has improved model robustness; however, further evaluation is required against more sophisticated evasion tactics as the landscape of adversarial attacks evolves.

In future, integrating semi-supervised learning could enhance real-time adaptation, particularly in environments where labelled data is scarce. Another key area for improvement is computational efficiency—while HDDAF optimises retraining frequency, further enhancements are required to ensure minimal latency in deployment. Explainability is another aspect that requires attention; integrating advanced XAI techniques will help security analysts better interpret HDDAF decisions, increasing its trustworthiness in critical applications. Finally, expanding the application beyond cybersecurity to fields, such as healthcare anomaly detection and industrial monitoring, will provide broader validation of the adaptability of such an approach to diverse, dynamic environments.

Overall, HDDAF represents a significant advancement in AI-driven anomaly detection, demonstrating that a hybrid approach that integrates multiple drift management strategies can offer both high accuracy and efficiency. However, practical deployment challenges remain, particularly in optimising computational costs and ensuring continued robustness against emerging threats.

Data Availability The data that support the findings of this study are openly available in the following public repositories:

- CIC-IDS2017 dataset at the Canadian Institute for Cybersecurity, University of New Brunswick, available at `https://www.unb.ca/cic/datasets/ids-2017.html`
- UNSW-NB15 dataset at the University of New South Wales, available at `https://research.unsw.edu.au/projects/unsw-nb15-dataset`

Declaration of Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments. This publication is part of the CiberIA, financed by "European Union NextGeneration-EU, the Recovery Plan, Transformation and Resilience, through INCIBE.

References

1. H. Abdi and L. J. Williams. Principal component analysis. *WIREs Computational Statistics*, 2(4):433–459, July 2010.

2. G. J. Aguiar and A. Cano. Enhancing concept drift detection in drifting and imbalanced data streams through meta-learning. In *2023 IEEE International Conference on Big Data (BigData)*. IEEE, Dec. 2023.
3. J. P. Barddal, H. M. Gomes, F. Enembreck, and B. Pfahringer. A survey on feature drift adaptation: Definition, benchmark, challenges and future directions. *Journal of Systems and Software*, 127:278–294, 2017.
4. N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. Smote: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, June 2002.
5. K. A. ElDahshan, A. A. AlHabshy, and B. I. Hameed. Meta-heuristic optimization algorithm-based hierarchical intrusion detection system. *Computers*, 11(12):170, Nov. 2022.
6. R. Fan, M. Zhong, S. Wang, Y. Zhang, A. Andrew, M. Karagas, H. Chen, C. Amos, M. Xiong, and J. Moore. Entropy-based information gain approaches to detect and to characterize gene-gene and gene-environment interactions/correlations of complex diseases. *Genetic Epidemiology*, 35(7):706–721, Oct. 2011.
7. A. Halbouni, T. S. Gunawan, M. H. Habaebi, M. Halbouni, M. Kartiwi, and R. Ahmad. Cnn-lstm: Hybrid deep neural network for network intrusion detection system. *IEEE Access*, 10:99837–99849, 2022.
8. R. Harwahyu, F. H. Erasmus Ndolu, and M. V. Overbeek. Three layer hybrid learning to improve intrusion detection system performance. *International Journal of Electrical and Computer Engineering (IJECE)*, 14(2):1691, Apr. 2024.
9. J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang. Learning under Concept Drift: A Review. *IEEE transactions on knowledge and data engineering*, page 1, 1 2018.
10. D. Mulimani, P. Patil, S. Totad, and R. Benni. Online detection and adaptation of concept drift in streaming data classification. *Procedia Computer Science*, 235:2803–2811, 2024.
11. B. Nelson, M. Barreno, F. J. Chi, A. D. Joseph, B. I. P. Rubinstein, U. Saini, C. Sutton, J. D. Tygar, and K. Xia. Exploiting machine learning to subvert your spam filter. In *Proceedings of the 1st Usenix Workshop on Large-Scale Exploits and Emergent Threats*, LEET'08, USA, 2008. USENIX Association.
12. M. S. Noori, R. K. Z. Sahbudin, A. Sali, and F. Hashim. Feature drift aware for intrusion detection system using developed variable length particle swarm optimization in data stream. *IEEE Access*, 11:128596–128617, 2023.
13. A. Pesaranghader and H. L. Viktor. *Fast Hoeffding Drift Detection Method for Evolving Data Streams*, page 96–111. Springer International Publishing, 2016.
14. M. Ring, D. Schlör, D. Landes, and A. Hotho. Flow-based network traffic generation using generative adversarial networks. *Computers and Security*, 82:156–172, May 2019.
15. I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani. Toward generating a new intrusion detection dataset and intrusion traffic characterization. In *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, pages 108–116, January 2018.
16. S. Sheikhi and P. Kostakos. A novel anomaly-based intrusion detection model using psogwo-optimized bp neural network and ga-based feature selection. *Sensors*, 22(23):9318, Nov. 2022.
17. A. V. Turukmane and R. Devendiran. M-multisvm: An efficient feature selection assisted network intrusion detection system using machine learning. *Computers and Security*, 137:103587, Feb. 2024.

18. T. Wisanwanichthan and M. Thammawichai. A double-layered hybrid approach for network intrusion detection system using combined naive bayes and svm. *IEEE Access*, 9:138432–138450, 2021.
19. D. Wishart. *k-Means Clustering with Outlier Detection, Mixed Variables and Missing Values*, page 216–226. Springer Berlin Heidelberg, 2003.
20. Q. Xiang, L. Zi, X. Cong, and Y. Wang. Concept Drift Adaptation Methods under the Deep Learning Framework: A Literature Review. *Applied Sciences*, 13(11):6515, 5 2023.
21. H. Zhang, X. Chen, M. Hu, and V. Sugumaran. Multilayer concept drift detection method based on model explainability. *IEEE Access*, page 1–1, 2024.
22. R. Zhao, Y. Mu, L. Zou, and X. Wen. A hybrid intrusion detection system based on feature selection and weighted stacking classifier. *IEEE Access*, 10:71414–71426, 2022.